

Taledge AI Interviewer: System Interaction Flow

The AI Interview Engine in Taledge is designed to conduct a dynamic, two-stage evaluation (Technical & Behavioural) mimicking a rigorous human interviewer. It relies on a Continuous Conversation Loop leveraging real-time STT (Speech-To-Text), LLM generation, and TTS (Text-To-Speech).

Here is the exact step-by-step breakdown of how the system interacts with the candidate.

1. Context Initialization (The Payload)

Before the interview begins, the frontend compiles the candidate's parsed resume and the target role into a strict JSON payload. This payload is embedded into the System Prompt.

Sample Context Payload:

```
{
  "candidate": {
    "name": "Alex Johnson",
    "targetRole": "Full Stack Developer",
    "resumeSkills": ["React", "Node.js", "PostgreSQL", "AWS"],
    "resumeProjects": [
      {
        "title": "E-Commerce Microservices",
        "impact": "Reduced load time by 40% and scaled to 10k daily active users."
      }
    ]
  },
  "stage": "Technical" // Can be "Technical" or "Behavioural"
}
```

2. The System Prompts (The Brain)

The AI is instructed strictly through role-based prompting depending on the current interview stage.

Stage 1: Technical Interview System Prompt

"You are an expert Technical Interviewer for Taledge interviewing the candidate for the role of {targetRole}. You possess lingual abilities to conduct the interview seamlessly in English, Hindi, and Hinglish. Adapt to the language the user speaks.

Your rules:

- 1. You must cross-examine the candidate strictly on the skills and projects mentioned in their resume.*
- 2. Ask highly technical follow-ups. If they mention 'Microservices', ask them about database partitioning or distributed tracing.*
- 3. Do not be overly friendly. Be rigorous, professional, and probing. Keep responses under 3 sentences to mimic natural conversation.*
- 4. Do not provide the answers if they fail. Move on to the next topic."*

Stage 2: Behavioural Interview (DNLA) System Prompt

"You are an expert HR Behavioural Analyst. Your goal is to map the candidate's responses to the 17 DNLA competencies (e.g., Empathy, Resilience, Leadership).

Your rules:

1. *Focus on the STAR method (Situation, Task, Action, Result).*
2. *Ask situational questions: 'Tell me about a time your technical solution failed in production.'*
3. *Probe their emotional regulation and conflict resolution.*
4. *Keep your responses short and conversational."*

3. The Real-Time Interaction Loop

Once the context is set, the interview enters a continuous loop:

1. **Candidate Speaks:** The candidate speaks into the microphone. The browser's native `SpeechRecognition` API (Webkit) instantly transcribes this into text.
2. **Data Transmission:** The transcribed text is pushed to the `messages` array and sent to our backend `app/api/interview/voice/route.ts`.
3. **AI Generation:** The backend forwards the entire chat history (including the hidden System Prompts) to **OpenRouter** (our LLM router).
4. **AI Responds:** The LLM streams back a concise, probing response.
5. **Speech Synthesis:** The frontend receives the text and uses the browser's `SpeechSynthesis` API to vocalize the response back to the candidate in a natural voice.

4. The Interview Termination

The interview is terminated under two conditions:

- **Natural Completion:** The AI concludes the interview after sufficient data is gathered (usually 10-15 minutes per stage).
- **Proctoring Violation:** If the candidate switches windows/tabs 3 times, the `visibilitychange` event listener forcefully terminates the session, logs a cheating flag, and halts the AI loop.